



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

FULL MARKS: 40
DATE: 07.11.2022

Instructions:

- The Question Paper consists of Two Parts. Part-I consist of Multiple Choice and Application based questions.
- Part -II consists of Java Programs.
- The intended marks for each question are given in brackets [].
- This paper consists of four printed pages.

PART-I (20 marks)
[Attempt all question]

Question 1

Choose the correct option:-

[1×10=10]

- a. What does the following method do?**

```
public static void numbers(int []num)
{ for(int x=1;x<num.length;x++)
  num[x]=num[0]; }
```

- i. It copies what is in cell 0 into all other cells
- ii. It changes every element to the value of x
- iii. It copies 0 into every element
- iv. It puts the array into numerical ascending order

- b. Predict the output of the following code fragment?**

```
int[] a=new int[5];
```

```
a[0] =5;
```

```
a[1]=4;
```

```
a[2]=3;
```

```
System.out.println(a[0]+ a[1]+a[4]);
```

- i. 9
- ii. 5 3 1
- iii. 5 3 0
- iv. 8

- c. “DPSNEWTOWN”.substring(2,5) will result in**

- i. SNE
- ii. SNEW
- iii. PSN
- iv. PSNE

d. What will be the output of the following code snippet?

```
class array_output {  
    public static void main( ) {  
        int array_variable [] = new int[10];  
        for (int i = 0; i < 10; ++i) {  
            array_variable[i] = i;  
            System.out.print(array_variable[i] + " ");  
            i++; } } }
```

i. 0 2 4 6 8

ii. 1 3 5 7 9

iii. 0 1 2 3 4 5 6 7 8 9

iv. 1 2 3 4 5 6 7 8 9 10

e. Given an array : int num[]={15,25,35,45,55};

What is the size of the array in bytes?

i. 5

ii. 20

iii. 10

iv. 40

f. What will be the output of the following code snippet?

```
class String_demo {  
    public static void main( ) {  
        char copy[ ] = {'a', 'b', 'c'};  
        String s = new String(copy);  
        String s1 = "abcd";  
        int len1 = s1.length();  
        int len2 = s.length();  
        System.out.println(len1 + " " + len2);}}
```

i. 3 0

ii. 0 3

iii. 3 4

iv. 4 3

g. Choose the correct output:-

```
public class Test{  
    public static void main( ){  
        String s1= "SITHA";  
        String s2= "RAMA";  
        System.out.println(s1.charAt(0) > s2.charAt(0));}}
```

i.true

ii.false

iii. 0

iv Compilation error

h. What is the value returned by the function compareTo() if the invoking string is less than the string compared?

i. zero

ii. value less than zero

iii. Value greater than zero

iv. None

- i. What is the output of the following println statement?

```
public class px{  
    public static void main(){  
        String str1="Hello";  
        System.out.println(str1.indexOf('t'));}  
}
```

- i. false ii. 0 iii. 1 iv. -1
- j. String to primitive type conversion is possible by using
- i. Interpreter ii. Compile iii. Wrapper class iv. ASCII conversion

Question 2

[2× 5=10]

- a. Write down the syntax to perform the following tasks:

Store the string “my mother is very loving” in S1 and substitute all occurrences of the letter ‘m’ by letter ‘M’ in S2. Then extract the last 3 characters of S2 and display it.

- b. The following numbers 89,20,31,56,15 are required to be sorted using Bubble sorting in an ascending order. What will be the result after the second pass.

- c. Predict the output of the following:-

```
String s1="good";  
String s2="world matters";  
String str1=s2.substring(5).replace('t','n');  
String str2=s1.concat(str1);  
System.out.println(str1);  
System.out.println(str2);
```

- d. Predict the output:-

```
int n[]={1,2,3,5,7,9,13,16};  
double x= Math.pow(n[4],n[2]);  
double y=Math.sqrt(n[5] + n[7]);  
System.out.println(“the value of x” +x+” ”+ “the value of y” +y);
```

- e. State the output of the following program segment:

```
String str1="great"; String str2="minds";  
System.out.println(str1.substring(0,2).concat(str2.substring(1)));  
System.out.println("WH"+(str1.substring(2).toUpperCase()));
```

PART-II (20 marks)

[Attempt all questions. Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted]

Question 3

[10]

Write a program in Java to accept 5 different cities names in a SDA. Arrange the names in ascending order using Bubble sort method and display them.

Sample Input: Enter the names of 5 cities

Delhi, Bangalore, Agra, Mumbai, Calcutta

Sample Output: The names arranged in ascending order are

Agra, Bangalore, Calcutta, Delhi, Mumbai

Question 4

Design a class *convert* with the following specifications:-

Data Members : String str

Member Methods

String reverse() :to return the reverse of the String. Reversing a string is the technique that reverses or changes the order of a given string so that the last character of the string becomes the first character of the string and so on. For example, we enter a string "APPLE", and then use the reverse algorithm. The reverse algorithm returns the string "ELPPA".

void toggle() :calls the reverse method and converts the reversed string to toggle case. Toggle Case capitalizes the small letters and changes the capital letter to small.

For instance, the word "dicTionary" will be converted into "DICtiONARY".

Write the main method to invoke the toggle method.